ECOLE MAROCAINE DES SCIENCES DE L'INGENIEUR
Membre de HONORIS UNITED UNIVERSITIES

MaintiDoc

Multimodal Maintenance Assistant with Verifiable Citations

HANFAOUI Karim[1]

[1]EMSI, Engineering Degree in Artificial Intelligence
Academic Supervisor: Mme. Kaoutar Karboub
Class: 5IADO-G2

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Role: Present the selected narrative and section order **Project Stage:** Navigation / global framing

1. Project Framing
2. Method and Planning
3. Design and Implementation
4. Validation and Results
5. Limits and Next Steps

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Problem Statement

Role: State the concrete need and the central question **Project Stage:** Scientific framing

Central Question

Can a technician photograph a control panel, ask what a warning light means, and receive an answer backed by the exact manual page — without the system ever guessing, and without it ever reading a document that technician is not allowed to see?

Current limitation: Maintenance knowledge is trapped in PDF manuals, scanned sheets and past intervention logs. A technician under pressure has no fast way to reach the right page, and a general chatbot will happily invent an answer with no source.

Expected outcome: A documentation assistant that retrieves from authorised manuals, reads the photograph, cites page-level evidence for every claim, and refuses to answer when the evidence is missing — while treating every document as data, never as instructions.

What MaintiDoc Delivers

Role: Define the four answer outcomes the system may produce **Project Stage:** Scientific framing

answered: A sourced answer, with at least one citation to a retrieved element. An answer without evidence is structurally impossible.

needs_clarification: The request is ambiguous or the photo is unreadable, so the system asks one precise question instead of guessing.

abstained: No accessible source covers the question. The system lists the available evidence and what is missing.

needs_approval: The user asked for a sensitive action — a ticket. The exact content is shown for human approval before anything is created.

Non-negotiable design rule

No model-invented confidence percentage exists anywhere in the contract. The output schema forbids unknown fields, so a fabricated score cannot even be represented.

Scope and Non-Goals

Role: Separate what the system is from what it must never be **Project Stage:** Scientific framing

In Scope

- Multimodal retrieval over authorised maintenance documents.
- Page-level citations that must support the claim they carry.
- Group-based access control on retrieval *and* on opening a file.
- Drafting a maintenance ticket, gated by explicit human approval.

Out of Scope by Design

- Controlling any equipment.
- Guaranteeing a diagnosis.
- Replacing safety procedures or a qualified technician.
- Acting autonomously: a human approves every sensitive action.

The system is a documentation aid. This boundary is stated in the product, the prompt and the tests — not only in the report.

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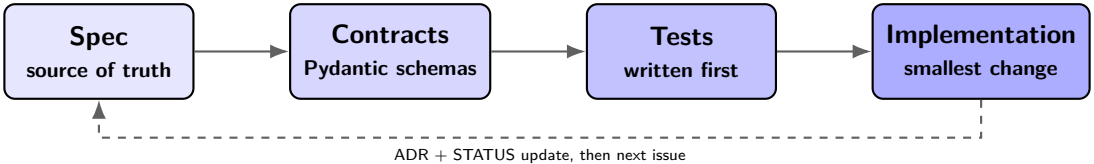
3 Design and Implementation

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Methodology

Role: Show how the work was organised and iterated **Project Stage:** Organisation / planning



Every decision that deviates from the specification is recorded in an Architecture Decision Record. The repository — not a chat history — is the memory between sessions.

Project Management — Task Table

Role: Summarise work packages, duration and real progress **Project Stage:** Organisation / planning

ID	Work package	Issues	State
W1	Framing, contracts, tooling, corpus	8	5 closed ✓
W2	Ingestion, embeddings, both indexes, access control	7	4 closed ✓
W3	VLM, citations, structured output, API/UI	6	1 closed
W4	MCP server, workflow, budgets, approval, tickets	5	not started
W5	Reranker dataset, fine-tuning, evaluation, security suites	6	not started
W6	Benchmarks, Docker deployment, demo, deliverables	5	not started
Total backlog		**37**	**10 closed (27 %)**

Work is tracked as GitHub Issues #1–#37 grouped into these six milestones.

Project Management — Gantt Chart

Role: Project the validated tasks onto a readable timeline **Project Stage:** Organisation / planning

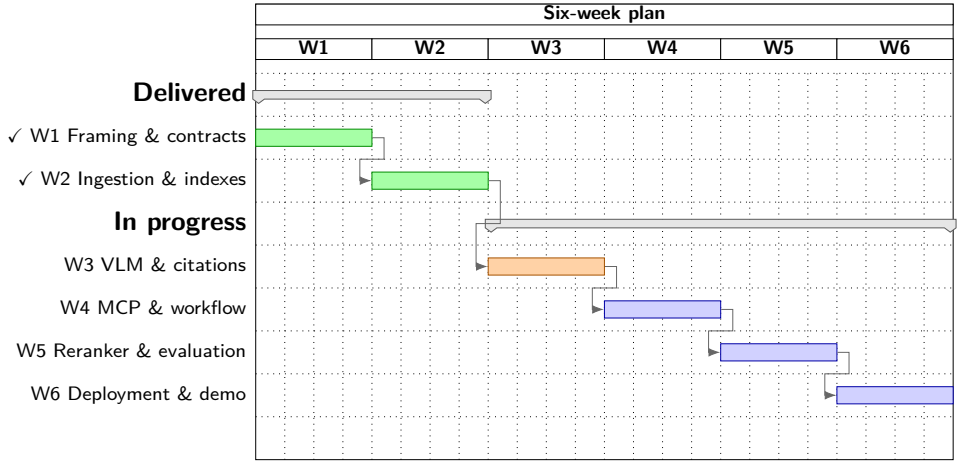


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Proposed Architecture

Role: Show boundaries, components, data flow and responsibilities **Project Stage:** Design

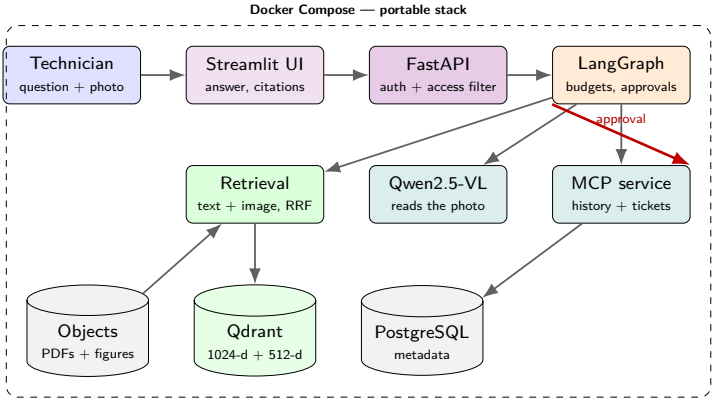


Figure 1: Component architecture. The access filter is applied inside the vector query, not after it.

Security Model — Three Hard Boundaries

Role: Explain the guarantees that are enforced by code, not by intent **Project Stage:** Design

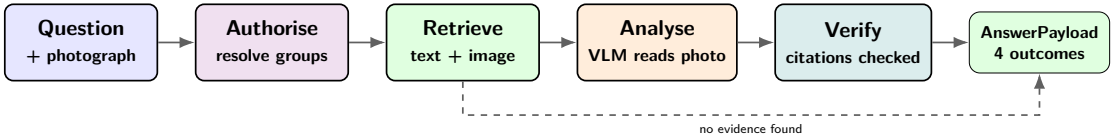
Boundary	How it is enforced	Proven by
User/group → documents	Filter applied *inside* the Qdrant query; deny by default, so an unlabelled document is visible to nobody	access-control tests
Model → sensitive action	Approval token is a signed JWT bound to the user *and* a SHA-256 of the exact content; single use, so a replay fails and any edit invalidates it	approval tests
Document text → instructions	Retrieved text is delimited and declared untrusted; the prompt loader *refuses to load* a prompt that does not delimit it	prompt-loader tests

Also enforced

A citation that was not actually retrieved, or that does not support its claim, cannot be produced: citations are built from the retrieved element itself. Opening a file checks permission *before* reading a single byte, and an unknown document is denied rather than reported as missing — otherwise it would be an enumeration oracle.

Core Workflow

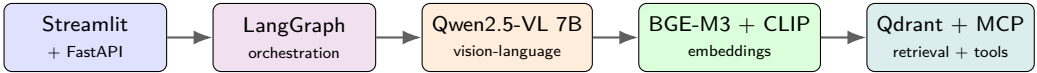
Role: Explain the most important end-to-end interaction **Project Stage:** Implementation



Budgets are enforced in code: at most 3 MCP calls, 6 model calls and 60 seconds per request. When the budget is exhausted the system abstains — it does not improvise.

Technology Choices

Role: Connect each technology to a concrete responsibility **Project Stage:** Implementation



Model choices

Qwen2.5-VL-7B-Instruct in 4-bit AWQ (fixed, never substituted) · BGE-M3, 1024-d multilingual text · CLIP ViT-B/32 image tower paired with a multilingual text tower, both licence-clean.

Infrastructure

Qdrant (vectors) · PostgreSQL (metadata) · object storage behind an interface · MLflow (runs) · Docker Compose, so the stack runs on a machine with no GPU.

What Was Actually Built

Role: Map the implemented modules to their responsibility **Project Stage:** Implementation

Module	Responsibility
`schemas`	Output contract: four statuses, citation model, status invariants, unknown fields rejected
`security`	Deny-by-default identity, vector-query filter, content-bound approval tokens, replay protection
`prompts`	Four versioned YAML prompts; loader rejects a prompt that does not delimit untrusted input
`ingestion`	PDF $\rightarrow$ paragraphs, tables, figures and OCR text, each with full document metadata
`retrieval`	BGE-M3 and CLIP embedders, two access-filtered indexes, RRF-ready rendering
`vlm`	Provider protocol plus a deterministic, citation-safe stub (the real model is next)
`runtime`	Device probe and a model ladder that explains every decision it makes
`storage`	Object-store interface: filesystem today, S3-compatible next
`corpus`	34 synthetic manuals, 2 equipment families, reproducible byte for byte

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Engineering Quality — The Gate

Role: Show the verification that every commit must pass **Project Stage:** Validation

Automatic checks

- **413 tests** pass, 41 of them against a live Qdrant container.
- **mypy –strict** clean on 32 modules.
- ruff lint and formatting clean.
- Secret scanning on every commit.

Tests that defend the promises

- An answer without a citation is rejected by the schema.
- A confidence field cannot exist: unknown fields are forbidden.
- A principal with no groups is *refused*, not given an empty filter.
- A denied file open reads **zero bytes** — asserted with a recording store.
- An approval token cannot be replayed, and editing the content invalidates it.

Documented divergences live in Architecture Decision Records: the task runner, the model ladder, the optional ML stack, the prompt format, object storage and the encoders.

Measured Model Behaviour

Role: Present verified measurements, including the uncomfortable one **Project Stage:** Validation

Check	Value	Verdict
BGE-M3 embedding width	1024-d	as configured
BGE-M3: related vs unrelated passage	0.849 vs 0.459	semantic ✓
CLIP image width	512-d	as configured
CLIP: figure vs itself / vs another figure	1.000 / 0.816	image search sound
CLIP text→image, *correct* panel	0.269	—
CLIP text→image, *wrong* panel	0.299	**wrong one wins**

Reported, not hidden

The last two rows are a real limitation: flat vector-drawn panels are far from the photographs CLIP was trained on, so cross-modal ranking does not hold on this corpus. Text→image quality is therefore *measured and reported separately* from image→image, and the vision-language model reading the actual figure remains the primary path. A number that flatters the project is worth less than one that is true.

Bugs Found by Executing, Not by Testing

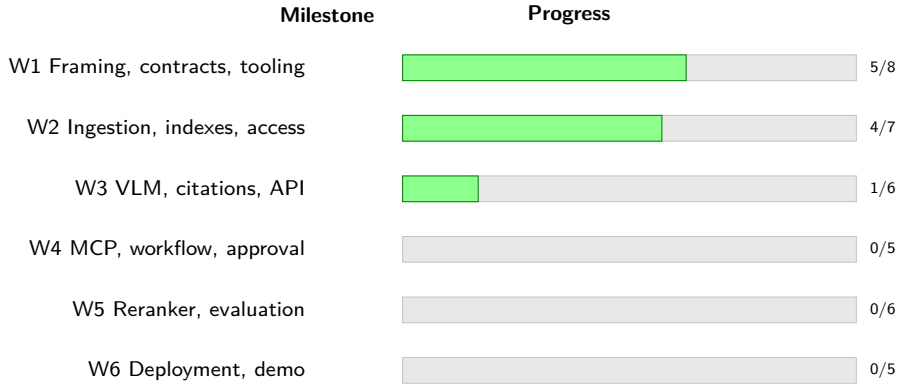
Role: Show that running the real stack found what doubles could not **Project Stage:** Validation

Symptom	Cause	Fix
Generated tables had no readable text at all	`insert_textbox` writes *nothing* when text overflows, returning a negative height	Measure and truncate, then draw with `insert_text`
Captions contained "?" instead of dashes	Built-in fonts cover Latin-1 only	Fail loudly on unrenderable characters
`Torch not compiled with CUDA enabled`	A visible GPU plus a CPU-only torch build	Mark the device *present but unusable*; never select it
Embedding call crashed on a missing tensor	A major library version changed the return type	Read the pooled output explicitly; verify against real weights

Each of these was invisible to unit tests written against stand-ins. They only surfaced when the real models and real PDFs ran — which is precisely why they are now recorded in the handoff document.

Project Progress

Role: Give an honest completion figure per milestone **Project Stage:** Validation



27 %
of the backlog closed (10/37)

≈30 %
by estimated effort

0 %
demoable end-to-end, for now

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Known Limits — Stated Plainly

Role: Separate what works from what has never run **Project Stage:** Closure

Works today

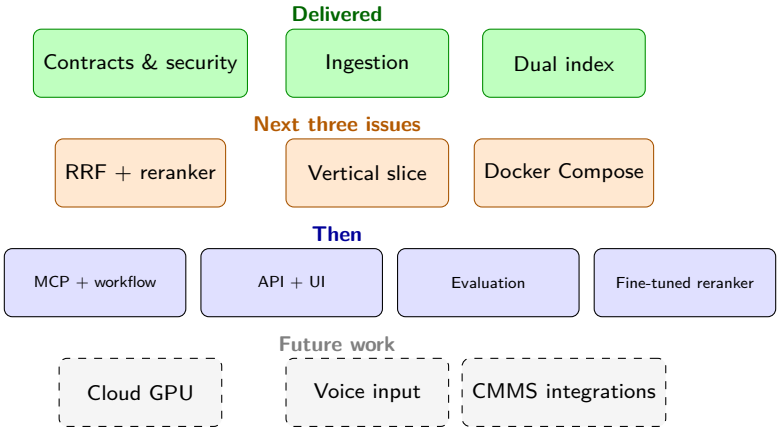
- Corpus generation: 34 documents, reproducible.
- PDF ingestion with tables, figures and OCR hooks.
- Real embeddings loaded and measured.
- Access-filtered search, verified against live Qdrant.
- Approval tokens with replay protection.

Not yet true

- **No question can be answered end to end yet.**
- The 7B vision-language model has never been loaded.
- OCR has never actually run (no Tesseract on the host).
- No Docker image or Compose file exists.
- No evaluation set, no fine-tuned reranker.

Roadmap

Role: Separate delivered scope from realistic future work **Project Stage:** Closure



Conclusion

Role: Close on the core value and the evidence **Project Stage:** Closure

Key Takeaway

MaintiDoc answers maintenance questions with evidence a technician can check — and refuses to answer when it has none. Integrity was treated as an engineering requirement: the schema cannot carry a fabricated confidence score, a citation cannot point at an unretrieved page, and a file cannot be read before permission is checked.

- `Contracts, security and retrieval core are built and verified: 413 tests.`
- `Real embeddings are loaded and measured, including a limitation that does not flatter the result.`
- `The remaining work is joined-up, not exploratory: workflow, tools, interfaces, deployment.`

Thank you — Questions?